

## Curriculum Vitae: updated October 4, 2026

### I. IDENTIFYING DATA

Name M. Brandon Westover, MD, PhD  
Current Position Emily Fisher Landau Professor of Neurology, Harvard Medical School, Beth Israel Deaconess Medical Center (BIDMC)  
Current Affiliation Beth Israel Deaconess Medical Center (BIDMC)

### II. EDUCATION HISTORY

#### Colleges and Universities Attended

7/1999 BS in Physics, Brigham Young University  
7/2006 PhD, Physics, Washington University in St. Louis  
7/2006 MD, Washington University School of Medicine

#### Residency and Fellowship Training

2006 – 2007 Internship, Barnes Jewish Hospital, Washington University School of Medicine, St. Louis, MO  
2007 – 2010 Residency, Partners Neurology Program (Massachusetts General Hospital and Brigham and Women's Hospital), Boston, MA  
2010 – 2012 Fellowship, Epilepsy and Clinical Neurophysiology, Massachusetts General Hospital, Boston, MA

#### Board Certification

2011 American Board of Psychiatry and Neurology (No. 56586)

### III. EMPLOYMENT

#### Academic Appointments:

07/12-05/14 Instructor in Neurology, Harvard Medical School, Boston, MA  
05/14-05/18 Assistant Professor in Neurology, Harvard Medical School, Boston, MA  
05/18-10/22 Associate Professor  
10/22 – present Professor of Neurology, Harvard Medical School, Boston, MA  
11/22 – present Emily Fisher Landau Professor of Neurology, Harvard Medical School, Boston, MA

#### Other Appointments:

2022 to present Director, Critical Care EEG Monitoring Service, BIDMC

### IV. HONORS AND AWARDS

2011 Best Paper Award Statistical Literacy ([www.statlit.org](http://www.statlit.org)) Statistical Education  
2012 Best Paper Award Association for Computing Machinery: Special Interest Group on Knowledge Discovery and Data Mining (SIGKDD) Basic Research  
2014 Rappaport Fellowship Massachusetts General Hospital Nominated award  
2018 #1 Most Disruptive AI Innovator Partners Healthcare Partners World Innovation Forum  
2018 Innovation Discovery Grant Partners Innovation Office Artificial Intelligence Research / Development  
2018 MGH-MIT Grand Challenge Winner ECOR Artificial Intelligence Research  
2018 Best Neurology Clinical Research Abstract MGH Neurology Department Clinical Research  
2018 Best Paper Award CyberWorlds 2018, Singapore Computer Science

|      |                                                                                                                                         |                                                                                                                                                                    |
|------|-----------------------------------------------------------------------------------------------------------------------------------------|--------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| 2019 | Disruptive Dozen AI Innovator                                                                                                           | Massachusetts General Hospital Partners HealthcareWorld Innovation Forum                                                                                           |
| 2019 | Innovation Discovery Award (IDG)                                                                                                        | Partners Healthcare World Innovation Forum                                                                                                                         |
| 2019 | Innovative Applications in Analytics Award (IAAA)                                                                                       | INFORMS Conference on Business Analytics & Operations ResearchFor achievements featuring novel combinations of descriptive, predictive and prescriptive analytics. |
| 2019 | Best Oral Abstract: Translational American Delirium Society (ADS)                                                                       | ADS annual meeting                                                                                                                                                 |
| 2019 | Derek Denny-Brown Young Neurological Scholar Award                                                                                      | American Neurological Association                                                                                                                                  |
|      | ANA's Highest award. For outstanding scientific advances toward the prevention, diagnosis, treatment, and cure of neurological diseases |                                                                                                                                                                    |

## V. BIBLIOGRAPHY

### Peer-reviewed original research (370 total)

1. Eliasmith C, **Westover MB**, Anderson CH. "A general framework for neurobiological modeling: an application to the vestibular system." *Neurocomputing* (Amst). 2002 Jun;44-46:1071-6. doi: 10.1016/s0925-2312(02)00418-6. PMCID: PMC6788744.
2. **Westover MB**, Eliasmith C, Anderson CH. "Linearly decodable functions from neural population codes." *Neurocomputing* (Amst). 2002 Jun;44-46:691-696. doi: 10.1016/s0925-2312(02)00459-9. Epub 2002 Mar 26. PMCID: PMC7062372.
3. Fischer BJ, **Westover MB**. "The neural multiple access channel." *Neurocomputing* (Amst). 2003 Jun;52-54:511-518. doi: 10.1016/s0925-2312(02)00762-2. Epub 2003 Jan 29. PMCID: PMC7062373.
4. **Westover MB**, Anderson CH. "Layer 4C in monkey V1 may linearize the output of the LGN." *Neurocomputing* (Amst). 2003 Jun;52-54:671-676. doi: 10.1016/s0925-2312(02)00863-9. Epub 2003 Jan 28. PMCID: PMC7062374.
5. **Westover MB**, O'Sullivan JA. "Achievable Rates for Pattern Recognition." *IEEE Trans Inf Theory*. 2008 Jan;54(1):299-320. doi: 10.1109/tit.2007.911296. Epub 2008 Jan 04. PMCID: PMC7062371.
6. **Westover MB**. "Asymptotic Geometry of Multiple Hypothesis Testing." *IEEE Trans Inf Theory*. 2008 Jul;54(7):3327-3329. doi: 10.1109/TIT.2008.924656. Epub 2008 Jun 17. PMCID: PMC6788803.
7. Woeltje KF, Butler AM, Goris AJ, Tutlam NT, Doherty JA, **Westover MB**, Ferris V, Bailey TC. "Automated surveillance for central line-associated bloodstream infection in intensive care units." *Infect Control Hosp Epidemiol*. 2008 Sep;29(9):842-6. doi: 10.1086/590261. PMCID: PMC6788749.
8. Mueen A, Keogh E, Zhu Q, Cash S, **Westover B**. "Exact Discovery of Time Series Motifs." *Proc SIAM Int Conf Data Min*. 2009;2009:473-484. doi: 10.1137/1.9781611972795.41. PMCID: PMC6814436.
9. Chu-Shore J, **Westover MB**, Bianchi MT. "Power law versus exponential state transition dynamics: application to sleep-wake architecture." *PLoS One*. 2010 Dec 02;5(12):e14204. doi: 10.1371/journal.pone.0014204. PMCID: PMC2996311.
10. Mueen A, Keogh E, Zhu Q, Cash SS, **Westover MB**, Bigdely-Shamlo N. "A disk-aware algorithm for time series motif discovery." *Data Min Knowl Discov*. 2011 Jan;22(1-2):73-105. doi: 10.1007/s10618-010-0176-8. Epub 2010 Apr 18. PMCID: PMC7062370.
11. Zhang WJ, **Westover MB**, Keary CJ. "Premortem diagnosis of sporadic Creutzfeldt-Jakob disease aided by positron-emission tomography imaging." *AJNR Am J Neuroradiol*. 2011 Jan;32(1):E18. doi: 10.3174/ajnr.A2292. Epub 2010 Nov 11. PMCID: PMC6788750.
12. **Westover MB**, Bianchi MT, Eckman MH, Greenberg SM. "Statin use following intracerebral hemorrhage: a decision analysis." *Arch Neurol*. 2011 May;68(5):573-9. doi: 10.1001/archneurol.2010.356. Epub 2011 Jan 10. PMCID: PMC3158138.
  - Featured article with accompanying editorial: *Arch Neurol*. 2011;68(5):565-566. "Statins After Intracerebral Hemorrhage: To Treat or Not to Treat".
  - Covered in over 150 news feeds, including CBS News, ABC News, US News and World Report, and Medscape Medical News.

13. **Westover MB, Westover KD**, Bianchi MT. "Significance testing as perverse probabilistic reasoning." BMC Med. 2011 Feb 28;9:20. doi: 10.1186/1741-7015-9-20. PMCID: PMC3058025.
14. **Westover KD, Westover MB**, Winer EP, Richardson AL, Iglehart JD, Punglia RS. "Should a sentinel node biopsy be performed in patients with high-risk breast cancer?." Int J Breast Cancer. 2011;2011:973245. doi: 10.4061/2011/973245. Epub 2011 Aug 10. PMCID: PMC3262582.
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 • Best Paper Award at Association for Computing Machinery's annual SIGKDD (Special Interest Group on Knowledge Discovery and Data Mining) meeting, 2012.
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- resource-limited settings: a decision analysis.” *Neurology*. 2014 Aug 26;83(9):787-93. doi: 10.1212/WNL.0000000000000730. Epub 2014 Jul 23. PMID: PMC4155044.
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#### **Peer-reviewed publications (accepted, in press)**

1. Nascimento FA, Katyal R, Kass NR, Yuan D, Sirven JI, Westover MB, Beniczky S. "Electroencephalography in clinical practice: results from an AAN Practice Current survey." *Journal of Clinical Neurophysiology*.\*\*
2. Sartipi S, Godfrey DS, Tauţan AM, Fernandes MP, Ghanta M, Gupta A, Nearing B, Kim J, Struck AF, Loddenkemper T, Peters J, Lee JW, Westover MB, Zafar SF. Automated Extraction of Seizures and Ictal-Interictal Continuum Patterns from EEG Reports to Enable Large-Scale Neurophysiology and Neurocritical Care Research. Available at SSRN. doi: 10.2139/ssrn.5489575. SSRN.\*\*

#### **Peer-reviewed publications (other - 8 total)**

1. Westover MB. "Review of Peter Dayan & L. F. Abbott's Theoretical Neuroscience: Computational and Mathematical Modeling of Neural Systems." *Philosophical Psychology*. 2002. 15: 563-567.
2. Bianchi MT and Westover MB. (2009) "Comment on: Analysis of SPARCL trial data for reduction of future stroke risk in older stroke patients with recent stroke or TIA." *Neurology* 73(10):817-818.
3. Westover MB, Gaspard N. "Not a simple plumbing problem: Updating our understanding of delayed cerebral ischemia in aneurysmal subarachnoid hemorrhage." *J Clin Neurophysiol*. 2016 Jun;33(3):171-3. PMCID: PMC4894327. Invited review as Guest Editor for a special issue of JCN on the neurophysiology of subarachnoid hemorrhage.
4. Dreier J, Fabricius M, Cenk A, Sakowitz O, Shuttleworth C, Dohmen C, Graf R, Vajkoczy P, Helbok R, Suzuki M; Schiefecker A.; Major S, Winkler M, Kang EJ, Milakara D, Oliveira-Ferreira A, Reiffurth C, Revankar G, Sugimoto K, Dengler N, Hecht N, Foreman B, Feyen B, Kondziella D, Friberg C, Piilgaard H, Rosenthal ES, Westover MB, Maslarova A, Santos E, Hertle D, Sanchez-Porrás R, Jewell S, Balança B, Platz J, Hinzman J, Luckl J, Schoknecht K, Schöll M, Drenckhahn C, Feuerstein D, Eriksen N, Horst V, Bretz J, Jahnke P, Scheel M, Böhner G, Rostrup E, Pakkenberg B, Heinemann U, Claassen J, Carlson A, Kowoll C, Lublinsky S, Chassidim Y, Shelef I, Friedman A, Brinker G, Reiner M, Kirov S, Andrew R, Farkas E, Guerresir, E, Vatter H, Chung L, Brennan K, Lieutaud T, Marinesco S, Maas A, Sahuquillo J, Dahlem M, Richter F, Herreras O, Boutelle M, Okonkwo D, Bullock R, Witte OW, Martus P, Van den Maagdenberg A, Ferrari M, Dijkhuizen R, Shutter L, Andaluz N, Schulte A, MacVicar B, Watanabe T, Woitzik J, Lauritzen M, Strong A, Hartings J. "Recording, analysis, and interpretation of spreading depolarizations in neurointensive care: review and recommendations of the COSBID research group." *Journal of Cerebral Blood Flow and Metabolism*. 2016 Jun 17. pii: 0271678X16654496. [Epub ahead of print] PMID: 27317657.
5. Westover MB, Halford JJ, Bianchi MT. What it should mean for an algorithm to pass a statistical Turing test for detection of epileptiform discharges. *Clin Neurophysiol*. 2017 Jul;128(7):1406-1407. doi: 10.1016/j.clinph.2017.02.026. Epub 2017 Mar 16. PMID: 28495216; PMCID: PMC9359665.
6. Bianchi MT, Thomas RJ, Westover MB. An open request to epidemiologists: please stop querying self-reported sleep duration. *Sleep Med*. 2017 Jul;35:92-93. doi: 10.1016/j.sleep.2017.02.001. Epub 2017 Feb 17. PMCID: PMC9371612.

7. Singla S, Garcia GE, Rovenolt GE, Soto AL, Gilmore EJ, Hirsch LJ, Blumenfeld H, Sheth KN, Omay SB, Struck AF, Westover MB, Kim JA. "Detecting Seizures and Epileptiform Abnormalities in Acute Brain Injury." *Curr Neurol Neurosci Rep*. 2020 Jul 27;20(9):42. doi: 10.1007/s11910-020-01060-4. Review. PMC9353690.
8. Nasiri S, Ganglberger W, Sun H, Thomas RJ, Westover MB. Exploiting Labels from Multiple Experts in Automated Sleep Scoring. *Sleep*. 2023 Feb 16;zsad034. doi: 10.1093/sleep/zsad034. Epub ahead of print. PMID: 36795078.

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#### **Non-Peer-reviewed Articles**

1. Westover MB, O'Sullivan JA. "Towards an information theoretic framework for object recognition." IEEE 2004 International Symposium on Information Theory, Chicago, Illinois, June 27-July 2, 2004.
2. Westover MB, O'Sullivan JA. "Achievable rates for pattern recognition: Binary and Gaussian cases." IEEE 2005 International Symposium on Information Theory, Adelaide, Australia, Sept 4-9, 2005.
3. O'Sullivan JA, Singla N, and Westover MB. "Successive Refinement for Pattern Recognition," IEEE Information Theory Workshop, Punta del Este, Uruguay, Feb 2006.
4. Liddle S, Zhang R, Grover L, Khitrov M, Brown J, Cobb JP, Goldman J, Chou J, Westover MB, Yagoda D, Reisner A, "Safety Evaluation of a Medical Device Data System". International Conference of the IEEE Engineering in Medicine and Biology Society (EMBC'12), San Diego, California, USA on August 28 - September 1, 2012.
5. Bianchi MT, Thomas RJ, Westover MB. "Response." *Sleep Med*. 2017 Aug 3. pii: S1389-9457(17)30316-7. doi: 10.1016/j.sleep.2017.07.017. [Epub ahead of print] PMID: 2884338
6. Amorim E, Zheng WL, Ghassemi MM, Aghaeiaval M, Kandhare P, Karukonda V, Lee JW, Herman ST, Sivaraju A, Gaspard N, Hofmeijer J, van Putten MJAM, Sameni R, Reyna MA, Clifford GD, Westover MB. "The International Cardiac Arrest Research (I-CARE) Consortium Electroencephalography Database." *medRxiv*. 2023 Aug 28:2023.08.28.23294672. doi: 10.1101/2023.08.28.23294672. Preprint. PMID: 37693458.
7. Sun H, Ganglberger W, Westover MB, Thomas RJ. "Sleep as a Window into Brain Health: Clinical Implications from the Sleeping Brain Wave-Structure Associations." *SLEEP*. 2025 May. (Editorial).\*\*

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#### **Book Chapters**

1. Westover MB, Thomas TT, Greer DM, "Delirium" in *Pocket Neurology*, ed. Westover MB, Choi E, Awad KM, and Greer DM, Lippincott Williams & Wilkins, 2010, 86-89.
2. Westover MB, Choi E, Awad KM, Greer DM, Editors. *Pocket Neurology*. Lippincott Williams & Wilkins, 2010.
3. Westover MB, Peters JM, Bromfield EB, "Seizures and Other Spells" in *Pocket Neurology*, ed. Westover MB, Choi E, Awad KM, and Greer DM, Lippincott Williams & Wilkins, 2010, 57-79.
4. Westover MB, Xia, Z, Cho TA, "Meningitis, Encephalitis, & Brain Abscesses" in *Pocket Neurology*, ed. Westover MB, Choi E, Awad KM, and Greer DM, Lippincott Williams & Wilkins, 2010, 136-149.
5. Thomas TT, Westover MB, "Rapidly Progressive Dementia" in *Pocket Neurology*, ed. Westover MB, Choi E, Awad KM, and Greer DM, Lippincott Williams & Wilkins, 2010, 89-96.
6. Kummer TT, Westover MB, "Central Nervous System Vasculitis" in *Pocket Neurology*, ed. Westover MB, Choi E, Awad KM, and Greer DM, Lippincott Williams & Wilkins, 2010, 175-177.
7. Westover MB, Awad K, "Ischemic Stroke, Muscular Dystrophy and Friedreich's Ataxia" in *MGH Cardiology Board Review*. Springer-Verlag, London, 2013.

8. Westover MB, Edlow BL, Greer DM, “Coma After Cardiac Arrest: Management and Neurological Prognostication” in MGH Cardiology Board Review. Springer-Verlag, London, 2013.
9. Westover MB, Shafi MM, “Seizures” in Pocket ICU, Lippincott Williams & Wilkins, 2013.

## **VI. GRANT FUNDING**

### **Current:**

|                   |                                                                                                                                                                                                                                                                               |
|-------------------|-------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| 09/2021 – 05/2026 | Funder: NIH-NINDS (R01 AG073410)<br>Title: Comparative Safety of Seizure Prophylaxis within the Medicare Program<br>Role: Co-Investigator (PI: Lidia Maria Moura)                                                                                                             |
| 05/2021 – 04/2026 | Funder: NIH-NINDS (R01 NS120947)<br>Title: Establishing a Brain Health Index from the Sleep Electroencephalogram<br>Role: PI                                                                                                                                                  |
| 09/2022 – 08/2027 | Funder: NIH-NHLBI (R01 HL161253)<br>Title: Data-Driven Sleep Biomarkers of Brain Health, Heart Health, and Mortality<br>Role: PI                                                                                                                                              |
| 08/2022 – 04/2027 | Funder: NIH-NIA (R01 AG073598)<br>Title: Low Neurophysiologic Resistance to Anesthetics as a Marker of Preclinical/Prodromal Alzheimer’s Disease and Neurovascular Pathology, Delirium Risk and Inattention<br>Role: Co-Investigator (PI: Miles Berger, via Duke subcontract) |
| 01/2023 – 12/2027 | Funder: NIH-NINDS (R01 NS126282)<br>Title: Prospective Validation of Neurophysiologic Outcome Prediction in Acute Brain Injury<br>Role: PI                                                                                                                                    |
| 09/2023 – 05/2028 | Funder: NIH-NINDS (R01 NS130119)<br>Title: Late-Onset Unexplained Epilepsy as a Risk Factor for Dementia<br>Role: Co-PI (with Alice D. Lam)                                                                                                                                   |
| 12/2023 – 11/2028 | Funder: NIH-NINDS (R01 NS131347)<br>Title: Comparative Effectiveness of EEG-Guided Anti-Seizure Treatment in Acute Brain Injury<br>Role: Co-Investigator (PI: Sahar F. Zafar)                                                                                                 |
| 11/2023 – 10/2028 | Funder: NIH-NIA (R01 AG078261)<br>Title: Delirium Dynamics: Understanding Causes and Effects<br>Role: Co-Investigator (PI: Eyal Kimchi, via Northwestern subcontract)                                                                                                         |
| 09/2023 – 09/2025 | Funder: NIH-NIA (R21 AG085495)<br>Title: Development of Novel Polysomnography-Based Digital Biomarkers to Predict Alzheimer’s Disease and Parkinson’s Disease<br>Role: Co-Investigator (PI: Yue Leng, via UCSF subcontract)                                                   |
| 02/2025 – 02/2026 | Funder: Takeda (PO8001791544)<br>Title: Analysis of Risk of Narcolepsy through Clinical Notes                                                                                                                                                                                 |

Role: PI

**Submitted:**

12/2025 – 11/2030

Funder: NIH (Pending)

Title: Leveraging Clinical Data and Automation to Predict Epilepsy After TBI

Role: Co-Investigator (PI: Jennifer Kim)

12/2025 – 11/2030

Funder: NIH (Pending)

Title: De-Biasing Effect Assessment of Seizure Effects on Neurologic Outcomes (DEBIAS)

Role: Co-Investigator (PI: Adrian Dalca)

12/2025 – 11/2030

Funder: NIH-NINDS (Pending)

Title: Automating Critical Care Electroencephalography Using AI Foundation Models

Role: PI

12/2027 – 11/2030

Funder: NIH (Pending)

Title: \*Tana Mu Na?\* Feasibility of Mobile EEG and Automated Intelligence to Diagnose Epilepsy in Children in Guinea-Bissau

Role: Co-Investigator (PI: Farrah Mateen)

07/2025 – 06/2030

Funder: NIH-NINDS (1UG3NS143749-01)

Title: Continuous Monitoring of Pain Using Facial Expressions in the ICU Environment (COMPANION)

Role: Co-Investigator (PI: Hao Deng)

**Completed:**

2012 – 2014

Funder: American Brain Foundation Clinical Research Training Fellowship

Title: Early Detection of Delayed Cerebral Ischemia After Subarachnoid Hemorrhage Using Context-Sensitive Computational EEG

Role: PI

2014 – 2019

Funder: NIH-NINDS (K23 NS090900)

Title: Quantitative Neurophysiological Monitoring and Control of Sedation-Analgesia in the ICU Environment

Role: PI

2017 – 2019

Funder: SAGE Therapeutics

Title: Automated Detection of Seizure-Spectrum Patterns and Quantitative Neurological Outcome Prediction

Role: PI

2018 – 2019

Funder: Partners Healthcare Innovation

Title: Artificial Intelligence-Based Seizure Detection and Classification

Role: PI

2018 – 2019

Funder: MGH-MIT Grand Challenge

Title: Big Data and Deep Learning for the Interictal-Ictal-Injury Continuum

Role: PI

- 2018 – 2021 Funder: Glenn Foundation for Medical Research  
Title: Electroencephalogram-Based Brain Age and Its Relationship with Cognitive Function and Sleep Quality  
Role: PI
- 2018 – 2023 Funder: NIH-NINDS (R01 NS102190)  
Title: Investigation of Sleep in the Intensive Care Unit  
Role: PI
- 2019 – 2024 Funder: NIH-NINDS (R01 NS107291)  
Title: Deep Learning for the Ictal-Interictal-Injury Continuum  
Role: PI
- 2018 – 2023 Funder: NIH-NINDS (R01 NS102574)  
Title: Multimodal Network Connectivity Architecture (MOCA) of the Brain and Its Role in Recovery of Consciousness in Comatose Cardiac Arrest Patients  
Role: Co-Investigator (PI: Ona Wu)
- 2019 – 2024 Funder: NIH-NIA (R01 AG060987)  
Title: Treating Hyperexcitability in Alzheimer’s Disease with Levetiracetam to Improve Brain Function and Cognition  
Role: Co-Investigator (PI: Mouhsin Shafi)
- 2019 – 2024 Funder: NIH-NIA (RF1 AG064312)  
Title: Integrated Motor Activity Biomarker for Risk of Alzheimer’s Dementia  
Role: Co-Investigator (PI: Kun Hu)

## **VII. PROFESSIONAL ACTIVITIES**

### **Editorial Service**

- 2010 Editor, Pocket Neurology, Lippincott Williams & Wilkins  
2016 Editor, Pocket Neurology, 2nd Edition; Wolters Kluwer  
2019-2024 Editorial Board Member, Journal of Clinical Neurophysiology

### **Ad Hoc Reviewer**

- 2010- present Neurology  
2010- present JAMA Neurology (formerly Archives of Neurology)  
2006- present Philosophical Psychiatry  
2010- present Journal of Clinical Epidemiology  
2010- present BMJ Case Reports  
2010- present IEEE Transactions on Information Theory  
2010- present IEEE Transactions on Signal Processing  
2010- present Journal of Medical Engineering  
2010- present Journal of Clinical Neurophysiology  
2010- present Journal of Neuroscience  
2010- present European Journal of Neurology  
2010- present Critical Care Medicine  
2010- present Annals of Clinical and Translational Neurology  
2010- present Journal of Neural Engineering  
2010- present Journal of Cerebral Blood Flow & Metabolism  
2010- present Computers in Biology and Medicine

|               |                                                            |
|---------------|------------------------------------------------------------|
| 2010- present | Proceedings of the National Academy of Sciences (PNAS)     |
| 2010- present | Brain                                                      |
| 2010- present | Epilepsy Research                                          |
| 2010- present | American Journal of Respiratory and Critical Care Medicine |
| 2010- present | SLEEP                                                      |

#### **Service as Grant Reviewer**

|            |                                                              |
|------------|--------------------------------------------------------------|
| 2018-2020  | Ad Hoc Member, Acute Neural Injury & Epilepsy (ANIE), NIH    |
| 2021- 2025 | Permanent Member, Acute Neural Injury & Epilepsy (ANIE), NIH |

#### **University Administrative Service**

|           |                                                                             |
|-----------|-----------------------------------------------------------------------------|
| 2011-2017 | Critical Care Informatics Committee Massachusetts General Hospital          |
| 2011-2022 | Neurocritical Care Clinical Review Committee Massachusetts General Hospital |
| 2011-2022 | Co-director, MGH Neuromonitoring Committee Massachusetts General Hospital   |
| 2010-2016 | Member, MGH EEG Reporting System Redesign Committee, MGH                    |
| 2013-2017 | MGH Infusion Pump Committee Massachusetts General Hospital                  |
| 2014-2022 | MGH Cooling Committee Massachusetts General Hospital                        |
| 2015-2018 | Member, PhD Thesis Committee (Candidate: Mohammad Ghassemi), MIT            |
| 2015-2019 | Member, PhD Thesis Committee (Candidate: Jing Zhi An), MIT                  |
| 2017-2019 | Chair, Data Safety Monitoring Board (DSMB) for R01 (PI: Ken Solt), MGH      |
| 2018-2020 | Member, DSMB for R01 (PI: Alvaro Pascual Leone), BIDMC                      |

#### **Service to Professional Organizations**

|           |                                                                                                                                                          |
|-----------|----------------------------------------------------------------------------------------------------------------------------------------------------------|
| 2011-2014 | TRENDs (Treatment of Recurrent Nonconvulsive SeizureS) Clinical Trial Investigators Committee, Duke University / Duke Clinical Research Institute (DCRI) |
| 2015-2017 | Member at Large Critical Care EEG Monitoring Research Consortium (CCEMRC)                                                                                |
| 2016      | Co-Chair, 7th International Conference on Extreme Learning Machines (ELM2016). Singapore, December 13-15, 2016.                                          |
| 2017-2019 | Vice-Chair, Critical Care EEG Monitoring Research Consortium (CCEMRC)                                                                                    |
| 2017      | Co-Chair, The 8th International Conference on Extreme Learning Machines (ELM2017). Yantai, China, October 4-7, 2017.                                     |
| 2019-2021 | Chair, Critical Care EEG Monitoring Research Consortium (CCEMRC)                                                                                         |
| 2025-2029 | Chair, International League Against Epilepsy (ILAE) – Commission on Big Data and AI                                                                      |
| 2025-     | Committee Member, NINDS Common Data Elements (CDE) in Epilepsy Research Oversight Committee.                                                             |

#### **Community Service**

|                                                   |                     |
|---------------------------------------------------|---------------------|
| 2012 Christ Lutheran Nursery School, Belmont, MA  | Educational Speaker |
| 2014 Boy Scout Troop 66, Belmont, MA. Belmont, MA | Educational Speaker |
| 2014 Butler Elementary School, Belmont, MA,       | Educational Speaker |
| 2016 Butler Elementary School, Belmont, MA,       | Educational Speaker |

### **VIII. INVITED PRESENTATIONS (SINCE 2020)**

#### **Grand Rounds:**

1. “Little; Yellow; Different: Hyperbilirubinemia in preterm infants of lactating depressed female Green Monkeys receiving CACKLE-modified dairy supplements.” Neonatology Grand Rounds, Stanford University, Stanford, CA (2022)

2. "Monitoring and Closed-Loop Control of Burst Suppression." Neurology Grand Rounds, Massachusetts General Hospital, Boston, MA (2013)
3. "Continuous Brain Monitoring in Critical Care." Anesthesia, Critical Care and Pain Medicine Grand Rounds, Massachusetts General Hospital, Boston, MA (2013)
4. "Why Brains Need Computers." Neurology Grand Rounds, Massachusetts General Hospital, Boston, MA (2016)
5. "Towards Personalized Critical Care Brain Monitoring." Neurology Grand Rounds, Stanford University, Palo Alto, CA (2017)
6. "Machine Learning in Neurology." Neurology Grand Rounds, University of Wisconsin, Madison, WI (2019)
7. "Early Ischemia Detection Using EEG in Subarachnoid Hemorrhage." Neurology Grand Rounds, University Hospitals Cleveland Medical Center, Cleveland, OH (2019)
8. "Automating Clinical Neurophysiology." Neurology Grand Rounds, Stanford University, Palo Alto, CA (2019)
9. "AI in Clinical Neurophysiology." Neurology Grand Rounds, Massachusetts General Hospital, Boston, MA (2020)
10. "Covid-19 Outcome Risk Prediction." Neurology Grand Rounds, Massachusetts General Hospital, Boston, MA (2020)
11. "Automated EEG Analysis in Clinical Care." Neurology Grand Rounds, University of California San Diego, San Diego, CA (2020)
12. "Denoising the Diagnosis of Epilepsy." Neurology Grand Rounds, Einstein Medical Center Philadelphia, Philadelphia, PA (2022)
13. "Quantifying the Ictal-Interictal-Injury Continuum." Neurocritical Care Grand Rounds, University of Washington, Seattle, WA (2022)
14. "Superhuman Management of Seizures in the ICU." Neurology Grand Rounds, Yale School of Medicine, New Haven, CT (2023)
15. "Optimizing Detection and Treatment of Nonconvulsive Status Epilepticus." Geschwind Pan-Harvard Grand Rounds, Harvard Medical School, Boston, MA (2023)
16. "Big Data and AI to Achieve Superhuman Detection and Treatment of Seizures in ICU Patients." Healthcare Delivery Science Grand Rounds, Beth Israel Deaconess Medical Center / Harvard Medical School, Boston, MA (2023)
17. "Toward Optimal Treatment of Harmful Brain Activity in Critical Illness." Neurology Grand Rounds, University Hospital Cleveland Medical Center, Cleveland, OH (2024)
18. "Reducing Noise in the Diagnosis of Epilepsy." Neurology Grand Rounds, University of Texas Houston, Houston, TX (2024)
19. "Reducing Noise in the Diagnosis of Epilepsy." Neurology Grand Rounds, University of Pennsylvania, Philadelphia, PA (2024)
20. "Reducing Diagnostic Noise in Neurology: Lessons from Epilepsy." Neurology Grand Rounds, Beth Israel Deaconess Medical Center / Harvard Medical School, Boston, MA (2025)
21. "Reducing Noise in Neurology: Lessons from Epilepsy." Neurology Grand Rounds, University of Michigan, Ann Arbor, MI (2025)

#### **Local Invited Presentations:**

1. "What is monkey cortex layer four for?." Neuroanatomy & Neurobiology Department, Washington University School of Medicine, St. Louis, MO (2002)
2. "EEG Patterns in the Neurological ICU." MGH Neurosciences ICU, Boston, MA (2010)
3. "Should Statins Be Avoided After Intracerebral Hemorrhage? Implications of SPARCL." MGH Stroke Research Center, Boston, MA (2010)
4. "Fundamentals of ICU EEG Monitoring." Nursing Skills Day, MGH Neurosciences ICU, Boston, MA (2012)

5. "Status Epilepticus and the Ictal–Interictal Continuum: Introduction to ICU EEG." MGH Critical Care Center, Boston, MA (2012)
6. "Neurological Prognostication in Coma After Cardiac Arrest." MGH Critical Care Center, Boston, MA (2013)
7. "Monitoring and Closed-Loop Control of Burst Suppression." Department of Brain & Cognitive Sciences, MIT, Cambridge, MA (2014)
8. "EEG-Related Skin Lesions: Statistical Process Monitoring & Risk Reduction." MGHfC Quality & Safety Operations Lead Team Meeting, Boston, MA (2014)
9. "Fundamentals of Continuous EEG Monitoring in Neurocritical Care; Quantitative Clinical EEG Fundamentals." MGH Neurosciences ICU & BWH/MGH Clinical Neurophysiology Fellows Didactic Conference, Boston, MA (2014)
10. "Automated Surveillance for Ventilator-Associated Pneumonia." MGH Infection Control Committee, Boston, MA (2015)
11. "The Role of Continuous EEG in Coma After Cardiac Arrest: Prognostic Value." MGH EEG Technologist Training Meeting, Boston, MA (2015)
12. "Quantitative Electroencephalography." Epilepsy Fellows Training Conference, MGH Neurology, Boston, MA (2017)
13. "CDAC: Bringing Big Clinical Data to Life." MGH Neurosurgery Faculty Meeting, Boston, MA (2018)
14. "Landmark Project #4: Clinical Time Series Data." Partners HealthCare EDMT Meeting, Boston, MA (2018)
15. "Validating the Brain Age Index: Modifying Brain Aging Through Exercise." McCance Center for Brain Health Faculty Meeting, Boston, MA (2019)
16. "Automatic Detection of Nonconvulsive Seizures." BIDMC Research Day, Boston, MA (2023)

#### **National and Regional Meetings:**

1. "Automatic Detection of Beta Amyloid in Digital Images Using Visual Texture." Alzheimer's Disease Research Center, Washington University, St. Louis, MO (2005)
2. "Early Detection of Delayed Cerebral Ischemia Using EEG." Greater Boston Epilepsy Society, Boston, MA (2015)
3. "Polysomnographic Signatures of Brain Age." Massachusetts Sleep Society, Westborough, MA (2017)
4. "Artificially Intelligent Brain Monitoring." World Medical Innovation Forum, Boston, MA (2019)
5. "Expert-Level Automatic Detection of Epileptiform Discharges." American Neurological Association, St. Louis, MO (2019)
6. "EEG Monitoring for Early Detection of Cerebral Ischemia." American Clinical Neurophysiology Society Annual Meeting, New Orleans, LA (2020)
7. "Automating Clinical Neurophysiology." Duke Comprehensive Epilepsy Center Quarterly Research Symposium, Durham, NC (2020)
8. "Big Data and Machine Learning to Advance Status Epilepticus Research." American Neurological Association, Virtual Meeting (2020)
9. "Automated Detection of Seizures and Ictal–Interictal Continuum Patterns Using Deep Learning." American Clinical Neurophysiology Society Annual Meeting, Virtual (2021)
10. "Sleep as a Window into Brain Health." Harvard Sleep Grand Rounds, Virtual (2021)
11. "Avoiding Pitfalls in EEG Interpretation." American Academy of Neurology Annual Meeting, Seattle, WA (2022)
12. "Artificial Intelligence in Sleep Medicine." American Neurological Association Annual Meeting, Chicago, IL (2022)
13. "Addressing Problems in Critical Care Neurophysiology Using Machine Learning." Machine Learning in Medicine Seminar Series, Carnegie Mellon University, Pittsburgh, PA (2022)
14. "Epileptiform and Non-Epileptiform Findings in ICU Electroencephalography." American Academy of Neurology Annual Meeting, Boston, MA (2023)

15. "Neurophysiological Grading of Encephalopathy." American Academy of Neurology Annual Meeting, Boston, MA (2023)
16. "Identifying and Overcoming Barriers for Clinical Use of Artificial Intelligence in Sleep Medicine." American Academy of Sleep Medicine Annual Meeting, Indianapolis, IN (2023)
17. "Optimizing Treatment of Nonconvulsive Status Epilepticus." American Epilepsy Society Annual Meeting, Orlando, FL (2023)
18. "Collaborating with Industry." American Clinical Neurophysiology Society Annual Meeting, Orlando, FL (2024)
19. "Quantitative EEG." American Clinical Neurophysiology Society Annual Meeting, Orlando, FL (2024)
20. "The EEG Map of Everything (MOE)." Critical Care EEG Monitoring Research Consortium Meeting, Orlando, FL (2024)
21. "AI in ICU EEG." American Epilepsy Society Annual Meeting, Los Angeles, CA (2024)
- 22.

### **International Meetings:**

1. "Opportunities for Intelligent Computing in Clinical Neurophysiology." Nanyang Technical University Symposium on Novel Technologies for Clinical Practice in Neurology, Singapore (2012)
2. "Real-Time Measurement and Closed-Loop Control of Burst Suppression." IEEE Engineering in Medicine and Biology Society Conference, Osaka, Japan (2013)
3. "Optimizing the Treatment of Refractory Status Epilepticus." Networks in Epilepsy Conference, Beijing, China (2015)
4. "Engineering in Neurocritical Care: Modeling & Control of Refractory Status Epilepticus." International Symposium on EEG Signal Analysis, Xi'an, China (2015)
5. "Quantitative Electroencephalography for Detection of Seizures and Ischemia." Beijing ICU Neuroelectrophysiological Monitoring Forum, Beijing, China (2015)
6. "Why Brains Need Computers: How Machine Learning and Big Data Can Help Neurology." Extreme Learning Machines Annual Conference, Singapore (2016)
7. "Machine Learning in Neurocritical Care." Northwestern University International Forum on Big Data in Healthcare, Xi'an, China (2017)
8. "Quantitative EEG: Detecting Ischemia and Seizures." International Congress of Clinical Neurophysiology, Washington, DC (2018)
9. "Removing the Confusion from Toxic-Metabolic Encephalopathy." University of Twente Mini-Symposium, Enschede, Netherlands (2018)
10. "Mapping the Ictal–Interictal–Injury Continuum." Neurocritical Care Society Annual Meeting, Vancouver, Canada (2019)
11. "Superhuman Detection of Seizures, Rhythmic and Periodic Patterns in ICU EEG." International Congress on Clinical Neurophysiology (ICCN), Geneva, Switzerland (2022)
12. "Quantifying the Ictal–Interictal Continuum." International Conference on Technology and Analysis for Seizures (ICTALS), Bern, Switzerland (2022)
13. "Denoising the Diagnosis of Epilepsy." AI in Epilepsy Conference, Breckenridge, CO (2025)
14. "MORGOTH: Automating the Interpretation of Clinical Neurophysiology Tests." International Conference on Technology and Analysis for Seizures (ICTALS), Montreal, Canada (2025)
15. "Data-Driven Biomarkers for Brain Health from Sleep." SLEEP 2025 Annual Meeting, Seattle, WA (2025)
16. "Artificial Intelligence in Neurology: Promise and Peril." Donald Baxter Lecture, Montreal Neurological Institute, Montreal, Canada (2025)

## **IX. TEACHING**

### **Teaching during Stanford appointment:**

None.

## **Teaching prior to Stanford:**

### **Courses**

|           |                                                                                                                                                                   |
|-----------|-------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| 2001–2002 | Instructor for Computational Neuroscience, Washington University School of Medicine, Department of Anatomy and Neurobiology.                                      |
| 2003      | Instructor for Detection and Estimation Theory, Washington University, Department of Electrical and Systems Engineering.                                          |
| 2004      | Instructor for Information Theory, Washington University, Department of Electrical and Systems Engineering.                                                       |
| 2008–2009 | Instructor for Neuroanatomy Laboratory, Harvard Medical School.                                                                                                   |
| 2012–2025 | Lecturer for Medical Decision Analysis and Diagnostic Test Interpretation (HST-192.0), Harvard–MIT Program in Health Sciences and Technology.                     |
| 2013–2017 | Course Director and Lecturer for Medical Decision Analysis and Diagnostic Test Interpretation (HST-192.0), Harvard–MIT Program in Health Sciences and Technology. |

### **Tutorials and Lectures**

|         |                                                                                                                                             |
|---------|---------------------------------------------------------------------------------------------------------------------------------------------|
| 10/2009 | Granger Causality Analysis Techniques for EEG Data. Clinical Neurophysiology Research Group, MGH, Boston, MA.                               |
| 5/2010  | Synchronization Likelihood for Functional Connectivity Analysis of EEG Data. Clinical Neurophysiology Research Group, MGH, Boston, MA.      |
| 7/2008  | Probability in Medical Reasoning. Neurology Residents, Brigham and Women's Hospital, Boston, MA.                                            |
| 9/2009  | Invited lecturer, Bayes' Rule for Physicians. Partners Neurology, MGH/BWH, Boston, MA.                                                      |
| 6/2010  | EEG Alphabet Soup: GCSE, NCSE, PLEDs, BIPLEDs, GPEDs, & DEADs. Partners Neurology, MGH/BWH, Boston, MA.                                     |
| 8/2011  | Status Epilepticus: Diagnosis and Management. Partners Neurology, MGH/BWH, Boston, MA.                                                      |
| 9/2012  | Introduction to ICU EEG: Seizures, Spikes, PEDs, Ischemia, and Coma. Partners Neurology, MGH/BWH, Boston, MA.                               |
| 7/2018  | Quantitative Electroencephalography. Neurophysiology and Epilepsy Fellows, MGH Epilepsy Division, Boston, MA.                               |
| 7/2019  | Recognizing Seizures and Ischemia in Critical Care EEG Monitoring. Neurophysiology and Epilepsy Fellows, MGH Epilepsy Division, Boston, MA. |
| 6/2020  | Emergency Electroencephalography. Emergency Medicine Residents, Massachusetts General Hospital, Boston, MA.                                 |
| 7/2020  | Introduction to Spectrograms in ICU EEG. Neurophysiology and Epilepsy Fellows, MGH Epilepsy Division, Boston, MA.                           |

### **Clinical Supervisory and Training**

|       |                                                                                                                                             |
|-------|---------------------------------------------------------------------------------------------------------------------------------------------|
| 2012– | Supervisor, MGH Adult Epilepsy Clinic (½-day session per week).                                                                             |
| 2012– | Attending Physician, MGH Inpatient Neurology Service (2-week block per year), supervising neurology residents and Harvard medical students. |
| 2012– | Attending Physician, MGH General Neurology Consult Service (2-week block per year), supervising neurology residents.                        |
| 2012– | Director, MGH Critical Care EEG Monitoring Service (≈10 weeks per year), supervising Epilepsy and Clinical Neurophysiology Fellows.         |
| 2012– | Attending, MGH Epilepsy Monitoring Unit (≈6 weeks per year), supervising Epilepsy and Clinical Neurophysiology Fellows.                     |
| 2012– | Faculty, Outpatient EEG Reading Service (≈2 days per month), supervising Epilepsy and Clinical Neurophysiology Fellows.                     |

### **Research Supervisory and Training Responsibilities**

2012– Supervisor of postdoctoral research fellows (average 5–10 fellows per year), Massachusetts General Hospital.

2012– Supervisor of graduate students (average 2–4 per year), Massachusetts General Hospital and MIT.

## **X. TRAINEES**

### **Postdoctoral Fellows**

2013-2014 Jing Jin, PhD from NTU in Singapore. postdoctoral fellow at MGH

2013-2014 Zhang Rui, PhD, visiting scholar from NTU in Singapore, mentoring for 2 months through present. Professor of Mathematics at Northwestern University, Xi'An, China

2014-2018 Sunil Belur Nagaraj, PhD, electrical engineering postdoctoral fellow. an Assistant Professor at University of Groningen, Department of Clinical Pharmacology, Groningen, Netherlands

2014- Lauren McClain, PhD. Research Fellow. a PhD student at Fordham University, New York

2019- Eline Oppersma, PhD. Visiting scholar. Assistant Professor of Technical Medicine at University of Twente, Netherlands

2019- WenDong Ge, PhD. Postdoctoral fellow.

2015–present Sahar Zafar, MD (K23).

2014–2020 Jennifer Kim, MD, PhD (K23).

2014–2019 Edilberto Amorim, MD (K23).

2019–present Marta Priscila Bento Fernandes, PhD.

2020–2022 Rajesh Amerineni, PhD.

2019–2022 Wendong Ge, PhD.

2019–2022 Weilong Zheng, PhD.

2022–2024 Yalda Amidi, PhD.

2021–2023 Samaneh Nasiri, PhD.

2019–present Wolfgang Ganglberger, PhD.

2024–present Shadi Sartipi, PhD.

2024–present Alexandra Tautan, PhD.

2024–present Ayush Tripathi, PhD.

2024–present Samuel Waters, PhD.

2024–present ChenXi Sun, PhD.

2024–present Charlotte Maschke, PhD.

2024–present Bruce Nearing, PhD.

2024–present Ayan Mitra, PhD.

2016–2018 Sungtae An, MS / PhD.

2016–present Haoqi Sun, MD.

2013–present Jing Jin, PhD.

### **Graduate Students**

2014- Mohammed Ghassemi, MIT PhD candidate (now graduated), mentorship resulted in Advanced Multimodal Neuroimaging Training Program (AMNTP) fellowship award, 3 published manuscripts. CEO of a startup company, Ghamut

2014- Jingzhi An, PhD candidate at Harvard/MIT HST program.

2014-2019 Moura, Junior Valdery, MS (computer science), programmer analyst / research assistant at MGH. Project: establishing a database for the Clinical Data Animation Center (CDAC). 3 peer-reviewed publications to d. Project Manager for Landmark Project #4 at Partners Healthcare

|              |                                                                                                                                                                |
|--------------|----------------------------------------------------------------------------------------------------------------------------------------------------------------|
| 2014-2017    | Apeksha Shenoy, MS (electrical engineering), programmer analyst / research assistant. a Senior Consultant at Boston Strategic Partners                         |
| 2015-        | Elham Bagheri, MS (electrical and systems engineering). Nanyang Technical University (NTU), Singapore, PhD candidate, co-supervised with Prof.                 |
| 2015-        | John Thomas, MS (electrical and systems engineering). Nanyang Technical University (NTU), Singapore, PhD candidate, co-supervised with Prof.                   |
| 2015-2016    | Siddharth Biswal, MS, computer programmer / research assistant. a PhD candidate at Georgia Institute of Technology                                             |
| 2015-        | Gladia Hotan, PhD candidate at MIT. Mentorship resulted in 2 publications to date.                                                                             |
| 2015-2016    | Ellis Wickering, MS, University of Twente, Netherlands. working as an computer programmer at Prolira                                                           |
| 2016-        | Jefferson T, Oliva, PhD candidate in University of Sao Paulo, Brazil, Institute of Mathematics and Computer Sciences.                                          |
| 2016-        | Jord Vink, MS, University of Twente, Netherlands, 3 month research internship with me at MGH, co-mentored with Dr. working at Brain Science Tools, Netherlands |
| 2016-        | Wei-Hung Sun, MD, masters student in the Harvard Biomedical Informatics program. a PhD candidate at MIT in computer science                                    |
| 2016-        | Manohar Ghanta, MS (computer science). Computer engineer / research assistant.                                                                                 |
| 2016-        | Yu-Ping Shao, MS (computer science). Computer engineer / data scientist / research assistant.                                                                  |
| 2016-        | Sungtae An, MS (computer science). Georgia Tech PhD candidate, co-supervised with Dr.                                                                          |
| 2016-        | Kunal Malhotra, MS (computer science). Georgia Tech (GAT) PhD candidate, co-supervised with Dr.                                                                |
| 2016-        | Haoqi Sun, MD (computer science). Nanyang Technical University (NTU), Singapore, PhD candidate, co-supervised with Prof.                                       |
| 2016-        | Senan Ebrahim, MD, PhD student in the Harvard Medical School HST program; co-supervised with Dr.                                                               |
| 2016-2017    | Sophia Bechek, research assistant. a PhD candidate at Stanford                                                                                                 |
| 2016-2018    | Balaji Goparaju, MSc. a Data Scientist at Novartis Institutes for BioMedical Research (NIBR)                                                                   |
| 2016-        | Jiang Ling Song, MS (mathematics), Northwestern University, Xian, China.                                                                                       |
| 2017         | Dennis Rebergen, Research internship.                                                                                                                          |
| 2017-        | Emile d'Angremont, Research internship.                                                                                                                        |
| 2017-2018    | Peiyun Ni, MD/PhD candidate at Harvard Medical School / MIT. Project: Antiepileptic drug treatment after an unprovoked first seizure: A decision analysis.     |
| 2016-2017    | Nishant Sinha, PhD candidate (now graduated) at NTU, Singapore, department of computer science.                                                                |
| 2018-        | Qiang Li, PhD candidate in Mathematics, Northwestern University, Xi'An, China.                                                                                 |
| 2018-2019    | Kicky van Leeuwen, Masters student in Technical Medicine from U. Twente, Netherlands. a Medical Machine Learning Scientist at Medix                            |
| 2018-        | Michael Leone, MSc.                                                                                                                                            |
| 2014-2020    | Jingzhi An, PhD.                                                                                                                                               |
| 2014-2019    | Mohammad Ghassemi, PhD.                                                                                                                                        |
| 2016-2017    | Jefferson T. Oliva, PhD.                                                                                                                                       |
| 2018-2020    | Qiang Li, PhD.                                                                                                                                                 |
| 2024-present | Arnav Gupta.                                                                                                                                                   |
| 2021-2025    | Thijs Nassi, MSc.                                                                                                                                              |

## Medical Students / Residents / Fellows

|           |                                                                                                                                                                                                                         |
|-----------|-------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
|           | Mentorship for 4 months (2 summers), resulted in 4 scientific publications.                                                                                                                                             |
| 2011-2012 | Sarah Wahlster, MD, Neurology resident. Assistant Professor of Neurology at University of Washington                                                                                                                    |
| 2011-2012 | Craig Williamson, MD, Neurology resident, Mentorship for 1 year, resulted in 1 published scientific manuscript. Assistant Professor of Neurology at University of Michigan                                              |
| 2013-2014 | Lidia Moura, MD, Epilepsy Fellow at MGH. Two abstracts, 1 published scientific manuscript. Assistant Professor at Harvard Medical School                                                                                |
| 2013-2014 | Marcus Ng, MD, Epilepsy Fellow at MGH. One published manuscript. Assistant Professor at University of Manitoba, Winnipeg, Canada                                                                                        |
| 2013-2014 | Deirdre O'Rourke, MD, Epilepsy Fellow at MGH, mentorship for 1 year, resulted in 1 abstract and 1 published manuscript. Neurologist at Connolly Hospital Blanchardstown/ Beaumont Hospital Dublin                       |
| 2013-2014 | Sandipan Pati, MD, Epilepsy Fellow at MGH. 1 abstract, 2 published scientific manuscripts. Assistant Professor of Neurology at UAB Birmingham                                                                           |
| 2013-2014 | Aaron Berkowitz, MD, PhD, neurology resident at BWH/MGH/Partners. Assistant Professor of Neurology at Harvard Medical School / Brigham and Women's hospital                                                             |
| 2014      | Vincent Alvarez, MD, visiting Research Fellow from Universite de Lausanne (Switzerland). 5 published manuscripts. an Attending Neurologist at Hopital du Valais, Sion, Switzerland                                      |
| 2014      | Sanjay Menon, MD, Neurosciences ICU Fellow at MGH, mentorship for 6 months.                                                                                                                                             |
| 2014      | Ammar Kheder, MD, Epilepsy Fellow at MGH, mentorship for 1 year. an Assistant Professor Of Clinical Neurology at U                                                                                                      |
| 2014-     | Aaron Struck, MD, Epilepsy Fellow at MGH. 2 abstracts, 8 published manuscripts. Assistant Professor of Neurology at University of Wisconsin                                                                             |
| 2014-2016 | Gina Deck, MD, Epilepsy Fellow at MGH, mentorship for 1 year. an Assistant Professor of neurology at The Warren Alpert Medical School of Brown University                                                               |
| 2014-     | Edilberto Amorim, MD, neurointensive care fellow at MGH. Research on the prognostic value quantification of EEG reactivity in coma in patients with anoxic brain injury.                                                |
| 2014-     | Jennifer Kim, MD, PhD, neurology resident at MGH/BWH. Project: "Identifying EEG Hallmarks for Increased Seizure Risk in SAH: A Quantitative EEG Analysis." Starting in July 2019, will be Assistant Professor of Neuro. |
| 2014-2017 | Katherine O'Connor, Project: Predicting delayed cerebral ischemia in patients with aneurysmal subarachnoid hemorrhage.                                                                                                  |
| 2015-     | Sahar Zafar, MD, neurointensive care fellow at MGH. Research on the impact of pathological brain activity and its treatments on neurological outcome in ICU patients.                                                   |
| 2015-2017 | Carlos Muniz, MD. Epilepsy / Neurophysiology Fellow. a practicing neurologist in Towson, Maryland                                                                                                                       |
| 2016-     | Ilya Toloch, MD, PhD, neurology resident at BWH/MGH/Partners.                                                                                                                                                           |
| 2016-2016 | Eva Postma, MD, University of Twente, Netherlands, visiting scholar for 6 months in my laboratory, co-mentored by Dr.                                                                                                   |
| 2016-2017 | Jia Jian, PhD (mathematics), Northwestern University (NWU), Xian, China. an Associate Professor of Mathematics at NWU                                                                                                   |
| 2016      | Honghui Zhang, PhD (mathematics), Northwestern University, Xian, China.                                                                                                                                                 |

|              |                                                                                                                                                                                                                                                                     |
|--------------|---------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| 2014-2018    | Durga Jonnalagadda, MD, Projects: Diagnosing severity of delirium in hospitalized patients from EEG recordings.                                                                                                                                                     |
| 2017-        | Aline Herlopian, MD. Fellow in Clinical Neurophysiology and Epilepsy at Harvard Medical School, MGH. Projects: Characterizing EEG patterns in patients with CAR-T cell encephalopathy, and Measuring expert inter-rater a. Assistant Professor of Neurology at Yale |
| 2017-        | Hassan Aboul Nour, MD. Project: Estimating the effect of exposure to inadvertent pharmacological burst suppression during critical illness on neurologic outcomes.                                                                                                  |
| 2017-2018    | Elahe Bordbar, MD. Research Fellow. a Resident in Neurology at Temple University                                                                                                                                                                                    |
| 2017-        | Muhammad Edhi, MD. Research Fellow.                                                                                                                                                                                                                                 |
| 2017-        | Farrukh Javed, MD. Research Fellow. a Neurology Resident at University of Virginia                                                                                                                                                                                  |
| 2017-        | Solomon Kassa, MD. Research Fellow.                                                                                                                                                                                                                                 |
| 2017-2019    | Maryum Shoukat, MD. Research Fellow.                                                                                                                                                                                                                                |
| 2017-        | Mohammad Tabaeizadeh, MD. Research Fellow.                                                                                                                                                                                                                          |
| 2017-2018    | Erik Bao, MD candidate in the Harvard/MIT Health Sciences & Technology.                                                                                                                                                                                             |
| 2017-2018    | Monica (Ling-Ya) Chao, MD candidate in Harvard HST program.                                                                                                                                                                                                         |
| 2018-        | Kendrick Shaw, MD, PhD. Anesthesia Resident and incoming faculty (Instructor) at Harvard Medical School / MGH. Project: Predicting delirium using electronic health record data.                                                                                    |
| 2018-        | Chris McGray, MD, PhD. Fellow in Epilepsy and Clinical Neurophysiology.                                                                                                                                                                                             |
| 2018-        | Sophia Ryan, MD. Harvard Medical School/CRICO Patient Safety and Quality Fellow at Harvard Medical School.                                                                                                                                                          |
| 2018-        | Benjamin Grannan, MD. Resident in Neurosurgery at Harvard Medical School / MGH. Project: Deep learning analysis of CT images to detect cases in need of surgical intervention.                                                                                      |
| 2019-        | Eliezer Sternberg, MD. Fellow in Epilepsy and Clinical Neurophysiology.                                                                                                                                                                                             |
| 2014–2015    | Elliott Akama-Garren.                                                                                                                                                                                                                                               |
| 2014–2020    | Jennifer Kim, MD, PhD.                                                                                                                                                                                                                                              |
| 2018–present | Eyal Kimchi, MD, PhD (K23).                                                                                                                                                                                                                                         |
| 2022–present | Daniel Goldenholz, MD, PhD (K23).                                                                                                                                                                                                                                   |
| 2022–present | Peter Hadar, MD, PhD.                                                                                                                                                                                                                                               |
| 2022–2025    | Wan Yee Kong, MD, MSc.                                                                                                                                                                                                                                              |
| 2022–present | James Lucarelli, MD, PhD.                                                                                                                                                                                                                                           |
| 2023–present | Callison Alcott, MD, PhD.                                                                                                                                                                                                                                           |
| 2011–2012    | Sarah Wahlster, MD.                                                                                                                                                                                                                                                 |
| 2011–2012    | Craig Williamson, MD.                                                                                                                                                                                                                                               |
| 2013–2014    | Lidia Moura, MD.                                                                                                                                                                                                                                                    |
| 2013–2014    | Marcus Ng, MD.                                                                                                                                                                                                                                                      |
| 2013–2014    | Patrick Chen, MD.                                                                                                                                                                                                                                                   |
| 2013–2014    | Deirdre O’Rourke, MD.                                                                                                                                                                                                                                               |
| 2013–2014    | Sandipan Pati, MD.                                                                                                                                                                                                                                                  |
| 2013–2014    | Ammar Kheder, MD.                                                                                                                                                                                                                                                   |
| 2013–2014    | Aaron Berkowitz, MD, PhD.                                                                                                                                                                                                                                           |
| 2017–2019    | Luis Paixao, MD.                                                                                                                                                                                                                                                    |
| 2017–present | Aline Herlopian, MD.                                                                                                                                                                                                                                                |
| 2020–2021    | Nitish Harid, MD.                                                                                                                                                                                                                                                   |

### Undergraduate Students

2010; 2011 Nathaniel Eiseman, Senior at Tulane.

|              |                                                                                                                                         |
|--------------|-----------------------------------------------------------------------------------------------------------------------------------------|
| 2012         | Matthew Ferguson, sophomore at Washington University in Saint Louis.                                                                    |
| 2012         | Justine Cormier, BS, medical student at Boston University School of Medicine.<br>Neurology Resident at Yale                             |
| 2014         | Elliot Akama-Garren, MIT undergraduate, mentorship for 6 months; resulted in 1 publication. an MD-PhD student at Harvard Medical School |
| 2016-        | Greta Gadbois, BS (mathematics), research assistant.                                                                                    |
| 2017         | Wade Whitt, undergraduate student at BYU. Projects: Correlation of EEG findings in delirious patients with neurologic outcomes.         |
| 2018-        | Jacob Hogan, undergraduate student at BYU. Project: Estimating the causal effect of burst suppression on neurologic outcomes.           |
| 2018–2023    | Alek Westover.                                                                                                                          |
| 2021–2022    | Danny Jones.                                                                                                                            |
| 2021–2022    | Jaden Barfuss.                                                                                                                          |
| 2021–2023    | Jayme Banks.                                                                                                                            |
| 2023–2024    | Leia Jekel.                                                                                                                             |
| 2023–2024    | Rebecca Milde.                                                                                                                          |
| 2023–2024    | Moritz Alkofer.                                                                                                                         |
| 2023–2024    | Elijah Davis.                                                                                                                           |
| 2023–present | Dan Sumsion.                                                                                                                            |
| 2023–present | Lydia Petersen.                                                                                                                         |
| 2023–present | Kiwi (Yiwen) Xiong.                                                                                                                     |
| 2024–present | Luke Jackson.                                                                                                                           |
| 2024–present | Carlos Nuñez-Sosa.                                                                                                                      |
| 2024–present | Amber Peng.                                                                                                                             |
| 2024–present | Mitchell McCauley.                                                                                                                      |
| 2024–present | Patrick Wynn.                                                                                                                           |
| 2024–present | Elijah Keldsen.                                                                                                                         |

### **High School Students**

|           |                                     |
|-----------|-------------------------------------|
| 2018      | Alek Westover, High school student. |
| 2016–2017 | Sophia Bechek.                      |
| 2023–2024 | Samvrit Rao.                        |